# Chapter 3: Discrete Random Variables and Probability Distributions

## LEARNING OBJECTIVES

1. Discrete random variables
2. Probability mass function and cumulative distribution function
3. Mean and Variance
4. Discrete Uniform Distribution
5. Binomial Distribution
6. Poisson Distribution

D.R. variables
PMF & CDF
Mean & Variance
Uniform Dis.
Binomial Dis.
Poisson Dis.
Summary

**Definition**

A **discrete** random variable is a random variable with a finite or countably infinite range.

**Example**

1. Roll a die twice: Let X be the number of times 4 comes up then X could be 0, 1, or 2 times.

2. Toss a coin 5 times: Let X be the number of heads then X = 0, 1, 2, 3, 4, or 5.

3. $X$ = The number of stocks in the Dow Jones Industrial Average that have share price increases on a given day then X is a discrete random variable because whose share price increases can be counted.

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## Determining a Discrete Random Variable

Let $X$ be a discrete random variable with possible outcomes $x_1, x_2, \ldots, x_n$.

1. Find the probability of each possible outcome.

2. Check that each probability is between 0 and 1 and that the sum is 1.

3. Summarizing results in following table:

| $X$ | $x_1$ | $x_2$ | .... | $x_n$ |
|---|---|---|---|---|
| $P(x)$ | $p_1$ | $p_2$ | .... | $p_n$ |

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**Example**

Let the random variable $X$ denote the number of heads in three tosses of a fair coin. Determine the probability distribution of $X$.

The sample space:

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$.

The events: $[X = 0] = \{TTT\}$ $\quad$ $[X = 1] = \{HTT, THT, TTH\}$

$[X = 2] = \{HHT, HTH, THH\}$ $\quad$ $[X = 3] = \{HHH\}$

| ***X*** | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| ***P(x)*** | 1/8 | 3/8 | 3/8 | 1/8 |

# Probability mass function

## Definition

For a discrete random variable $X$ with possible values $x_1$, $x_2$, …, $x_n$ , a **probability mass function** is a function such that

(1) $f(x_i) \geq 0$

(2) $\sum_{i=1}^{n} f(x_i) = 1$

(3) $f(x_i) = P(X = x_i)$

In above example, we have

$f(0) = 1/8$, $f(1) = 3/8$, $f(2) = 3/8$ and $f(3) = 1/8$.

# Cumulative distribution function



## Definition

The **cumulative distribution function** of a discrete random variable $X$, denoted as $F(x)$, is

$$F(x) = P(X \le x) = \sum_{x_i \le x} f(x_i)$$

For a discrete random variable $X$, F(x) satisfies the following properties.

(1) $0 \le F(x) \le 1$

(2) If $x \le y$, then $F(x) \le F(y)$

# Cumulative distribution function

## Example

Suppose that a day's production of 850 manufactured parts contains 50 parts that do not conform to customer requirements. Two parts are selected at random, without replacement, from the batch. Let the random variable $X$ equal the number of nonconforming parts in the sample.

What is the cumulative distribution function of $X$?

The first we find the probability mass function of $X$.

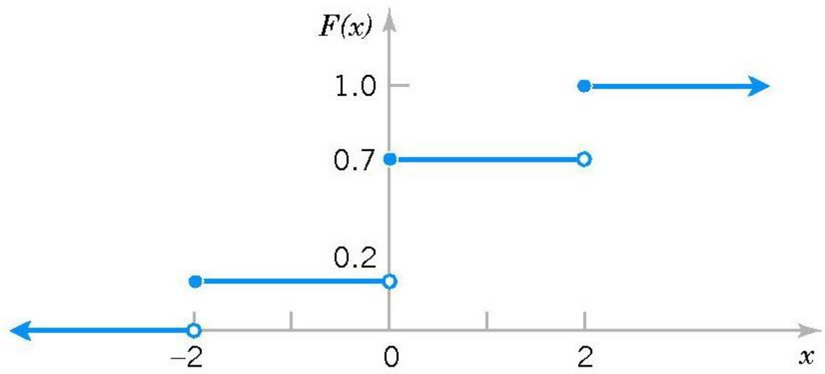
| $X$ | 0 | 1 | 2 |
|---|---|---|---|
| $f(x)$ | 0.886 | 0.111 | 0.003 |

$$\Longrightarrow F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 0.886 & \text{if } 0 \le x < 1 \\ 0.997 & \text{if } 1 \le x < 2 \\ 1 & \text{if } x \ge 2 \end{cases}$$

## Example

Determine the probability mass function of $X$ from the following cumulative distribution function:

$$F(x)=\begin{cases}0 & x<-2\\ 0.2 & -2\le x<0\\ 0.7 & 0\le x<2\\ 1 & x\ge 2\end{cases}$$




$f(-2)=0.2-0=0.2$

$f(0)=0.7-0.2=0.5$

$f(2)=1.0-0.7=0.3$

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## Definition

The **mean** or **expected value** of the discrete random variable $X$, denoted as $\mu$ or $E(X)$ is

$$\mu = E(X) = \sum_i x_i f(x_i)$$

The **variance** of $X$, denoted as $\sigma^2$ or $V(X)$ is

$$\sigma^2 = V(X) = E(X - \mu)^2$$

$$= \sum_i x_i^2 f(x_i) - \mu^2$$

The **standard deviation** of $X$ is $\sigma$.

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### Example

The number of messages sent per hour over a computer network has the following distribution:

| $X$ | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|
| $f(x)$ | 0.08 | 0.15 | 0.30 | 0.20 | 0.20 | 0.07 |

Determine the mean and standard deviation of the number of messages sent per hour.

$$E(X) = 10(0.08) + 11(0.15) + \cdots + 15(0.07) = 12.5$$

$$V(X) = 10^2(0.08) + 11^2(0.15) + \cdots + 15^2(0.07) - 12.5^2 = 1.85$$

$$\sigma = \sqrt{V(X)} = \sqrt{1.85} = 1.36$$

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## Mean of a Function of a Discrete Random Variable

If $X$ is a discrete random variable with probability mass function $f(x)$ then for any function $h(x)$

$$E[h(x)] = \sum_i h(x_i) f(x_i)$$

## Corollary

$$E(aX + b) = aE(X) + b$$

$$V(aX + b) = a^2 V(X)$$

# Discrete Uniform Distribution

## Definition

A random variable $X$ has a **discrete uniform distribution** if each of the $n$ values in its range, say, $x_1, x_2, \ldots, x_n$ has equal probability. Then,

$$f(x_i) = 1/n$$

## Mean and Variance

Suppose $X$ is a discrete uniform random variable on the consecutive integers $a, a+1, \ldots, b$ for $a \le b$. The mean and variance of $X$

$$\mu = E(X) = (a + b)/2 \qquad \sigma^2 = \frac{(b-a+1)^2 - 1}{12}$$

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## Definition

A random experiment consists of *n* Bernoulli trials such that

(1) The trials are independent

(2) Each trial results in only two possible outcomes, labeled as "success" and "failure"

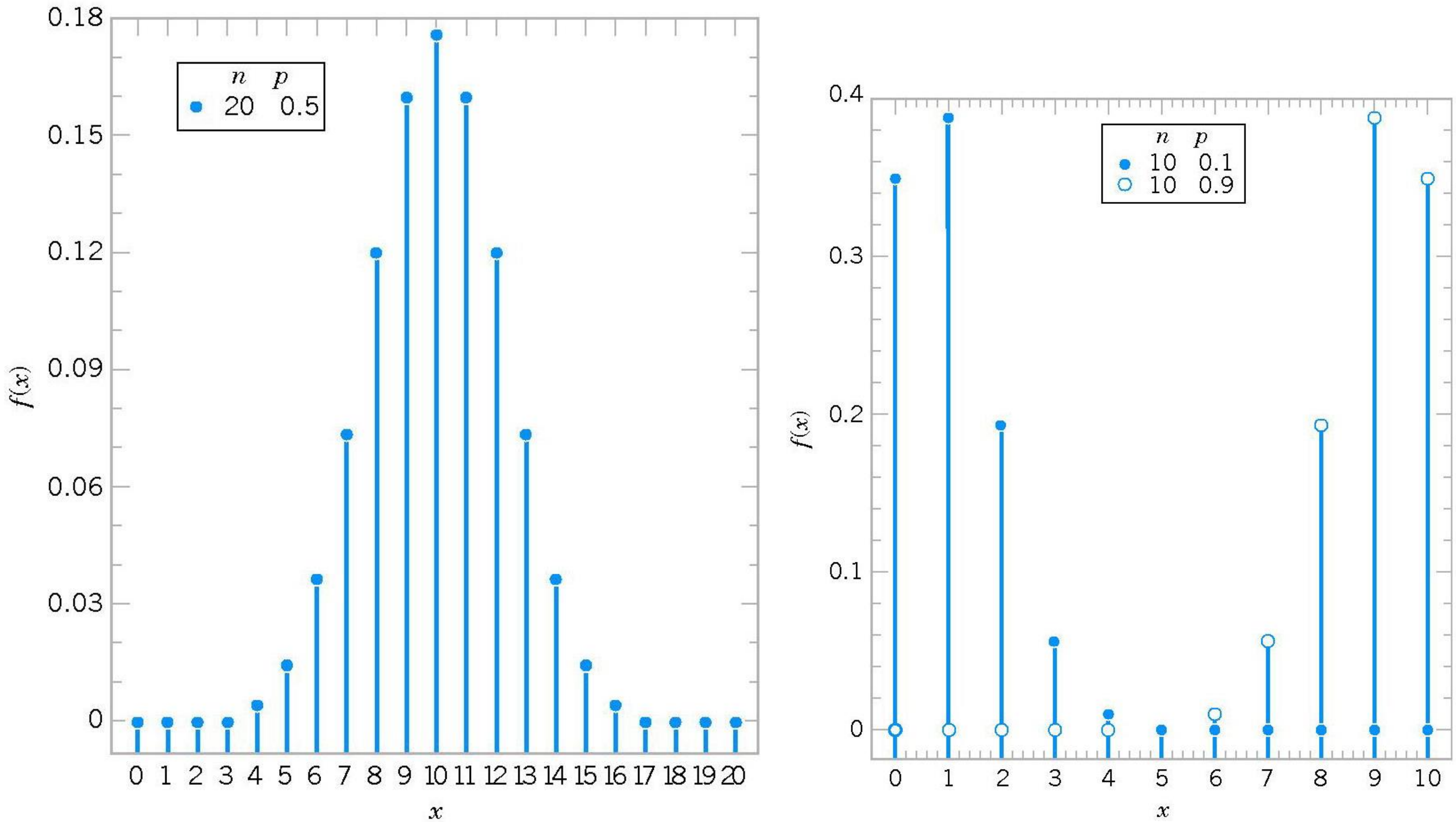
(3) The probability of a success in each trial, denoted as *p*, remains constant

The random variable *X* that equals the number of trials that result in a success has a **binomial random variable** with parameters $0<p<1$ and n=1,2,…The probability mass function of *X* is

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0, 1, 2, \ldots, n.$$

# Binomial Distribution






Binomial distributions for selected values of $n$ and $p$.

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## Mean and Variance

$\mu = E(X) = np$ $\sigma^2 = V(X) = np(1-p)$

### Example

Each sample of water has a 10% chance of containing a particular organic pollutant. Assume that the samples are independent with regard to the presence of the pollutant.

(a) Find the probability that in the next 18 samples, exactly 2 contain the pollutant.

(b) Determine the probability that at least four samples contain the pollutant.

(c) Determine the probability that $3 \le X < 7$.

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**Solution.** Let $X$ = the number of samples that contain the pollutant in the next 18 samples analyzed. Then $X$ is a binomial random variable with $p = 0.1$ and $n = 18$.

(a) $$P(x=2)=\binom{18}{2}0.1^2(1-0.1)^{18-2}=0.284$$

(b) $$P(x\ge 4)=\sum_{x\ge 4}\binom{18}{x}0.1^x(1-0.1)^{18-x}=0.098$$

(c) $$P(3\le x<7)=\sum_{x=3}^{6}\binom{18}{x}0.1^x(1-0.1)^{18-x}=0.265$$

# Poisson Distribution



## Definition

Given an interval of real numbers, assume events occur at random throughout the interval. If the interval can be partitioned into subintervals of small enough length such that

(1) the probability of more than one event in a subinterval is zero,

(2) the probability of one event in a subinterval is the same for all subintervals and proportional to the length of the subinterval, and

(3) the event in each subinterval is independent of other subintervals, the random experiment is called a **Poisson process.**

The random variable *X* that equals the number of events in the interval is a **Poisson random variable** with parameter $\lambda > 0$, and the probability mass function of *X* is

$$f(x) = e^{-\lambda}\frac{x^k}{x!} \qquad x = 0, 1, 2, \ldots$$

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## Mean and Variance

If $X$ is a Poisson random variable with parameter $\lambda$, then

$$\mu = E(X) = \lambda \qquad \sigma^2 = V(X) = \lambda$$

## Example

For the case of the thin copper wire, suppose that the number of flaws follows a Poisson distribution with a mean of 2.3 flaws per millimeter. Determine the probability of exactly 2 flaws in 1 millimeter of wire.

Determine the probability of at least 1 flaw in 2 millimeters of wire.

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Let $X$ denote the number of flaws in 2 millimeters of wire. Then, $X$ has a Poisson distribution with

$$E(X) = 2 \text{ mm} \times 2.3 \text{ flaws/mm} = 4.6 \text{ flaws}$$

Therefore,

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-4.6} = 0.9899$$

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Homework: Read slides of the next lecture.